gravity's intrinsic strength on small scales is far greater than previously thought, tiny black holes could be produced with significantly less compression force than previously believed. Calculations show that the Large Hadron Collider may have just enough squeezing power to create a cornucopia of microscopic black holes through high-energy collisions between protons.[7] Think about how amazing that would be. The Large Hadron Collider might turn out to be a factory for producing microscopic black holes! These black holes would be so small and would last for such a short time that they wouldn't pose us the slightest threat (years ago, Stephen Hawking showed that all black holes disintegrate via quantum processes—big ones very slowly, tiny ones very quickly), but their production would provide confirmation of some of the most exotic ideas ever contemplated.

Braneworld Cosmology

A primary goal of current research, one that is being hotly pursued by scientists worldwide (including me), is to formulate an understanding of cosmology that incorporates the new insights of string/M-theory. The reason is clear: not only does cosmology grapple with the big, gulp-in-the-throat questions, and not only have we come to realize that aspects of familiar experience—such as the arrow of time—are bound up with conditions at the universe's birth, but cosmology also provides a theorist with what New York provided Sinatra: a proving ground par excellence. If a theory can make it in the extreme conditions characteristic of the universe's earliest moments, it can make it anywhere.

As of today, cosmology according to string/M-theory is a work in progress, with researchers heading down two main pathways. The first and more conventional approach imagines that just as inflation provided a brief but profound front end to the standard big bang theory, string/M-THEORY provides a yet earlier and perhaps yet more profound front end to inflation. The vision is that string/M-theory will unfuzz the fuzzy patch we've used to denote our ignorance of the universe's earliest moments, and after that, the cosmological drama will unfold according to inflationary theory's remarkably successful script, recounted in earlier chapters.